JAKE GARRISON

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EDUCATION

University of Washington - Seattle, WA *Sept 2011 – Present*

B.S. Electrical Engineering (DSP and Communications), Math Minor, Computer science, Art and design coursework
M.S. Electrical Engineering (UbiComp Lab), signal processing and machine learning coursework

Specialties: DSP, Computer Vision, Machine learning, AI, Mobile/Web Dev, Audio Processing, Communications, Networking & Security, Power Electronics

RELEVANT EXPERIENCE

UbiComp Lab (https://ubicomplab.cs.washington.edu/) - Seattle, Washington *Sept 2016 – Present*

Ubiquitous computing research lab, led by Prof. Shwetak Patel focusing on mobile health sensing and novel interaction

- Using machine learning, sound and image processing on smartphones to screen and monitor health

Haiku Deck Developer (https://www.haikudeck.com/) - Seattle, Washington *June 2016 – Sept 2016*

A presentation iOS and Web app startup used by millions. It features Zuru, an AI platform for automation

- Full stack developer and R&D for Zuru and its new features. Wrote production ready Node.js and React.js code

Puppy.ai App - Seattle, Washington *Sept 2015 – Present*

Real-time dog breed detection AI using Tensorflow on iOS. One of three working remotely on project

- My duties include creating the classifier and iOS app design. App launches in Spring 2017

Urban Parking App (City of Seattle) - Seattle, Washington *Jan 2016 – Present*

Project to help route drivers to open parking spots based off historical data. Entrepreneurial capstone project

- Uses machine learning (boosting via python) to predict parking trends based on 45 million historical transactions

Driver Awareness System (https://uwecocar.github.io/UW-Infotainment/) - Seattle, Washington *July 2015 – June 2016*

Founded and lead a team designing a system and UI that monitors and logs metrics that contribute to bad driving

- Displays distraction, phone usage, sleep and happiness metrics using C++, Python, Node.js and D3.
- Won Most Innovative use of Data and Best Travel Hack in DubHacks and 3rd Place in EcoCAR Consumer Appeal

Tesla Motors Systems Integration – Palo Alto, California *June 2015 – Sept 2015*

Integration intern on a small team focused solely on the development of the Model X falcon doors and sensing

- Created production code and contributed to controls, firmware, sensing, processing, testing and validation

EcoCAR Competition Autopilot and Electrical Lead (http://www.ecocar3.org/) - Seattle, Washington *Sept 2011 – Present*

First Electrical Team Lead, now Assisted Driving (ADAS) Lead. Involved in EcoCAR 2 and 3 since freshman year

- Computer vision and autonomous vehicle controls research and simulation
- Embedded BSP Linux development with bash/python scripting and C/C++ programs
- Project management and collaboration with multidisciplinary students, faculty and industry

Bankroll Bitcoin Miner Startup - Palo Alto, California *June 2014 – Nov 2014*

- One of three involved, worked on PCB, firmware, schematic and enclosure design. Never launched

Tesla Motors Power Electronics - Palo Alto, California *June 2014 – Sept 2014*

Power Electronics intern focused on the high voltage electric powertrain for the Model S P85D Insane mode

- Testing and validation on existing parts (temperature and current cycling)
- R&D for future technology (specifically high voltage fuses, contactors and charging plugs)
- Creating professional CAD, drawings, schematics, and pcb designs

Electric Car Conversion (http://electricgti.blogspot.com) - Spokane, WA *Jan 2010 – May 2012*

Personally designed, funded and converted a gas powered Volkswagen GTI to fully electric

- Designed and hand built the high voltage motor controller along with several metal/wood components
- Awarded engineering scholarship, admission to UW and invited onto the team

Ion Beam Research Assistant – Seattle, Washington *May 2013 – Feb 2014*

Research Assistant for Professor Bruce Darling, UW Electrical Engineering

- Learned to simulate and design circuit boards using NI LabView Multisim and Ultiboard Work

Custom Music Equipment Designer (http://www.verellenamplifiers.com) - Seattle, Washington *July 2011 – 2014*

Personally designed, built and sold guitar pedals and other music gear

- AC and DC Circuit analysis and troubleshooting experience including tubes, transistors and amplifier circuits

SCHOLARSHIPS AND AWARDS

4.0 in multiple capstone projects (UW), 2nd place Autopilot and 3rd place Consumer appeal (EcoCAR 3) – San Deigo, CA *Spring 2016*
Most innovative use of data (from GE) and Best travel hack award (from Concur) at DubHacks – Seattle, WA *Fall 2015*
1st place Electrical presentation and 3rd place Mathworks presentation (EcoCAR 3) – Seattle, WA *Spring 2015*
2nd place Overall, Battery Pack Design and Electrical presentation (EcoCAR 2) - San Diego, CA *Spring 2013*
Washington Society of Engineers - Spokane, WA *Spring 2011*

AREAS OF TECHNICAL EXPERTISE

Platforms:	Windows, OSX, Linux, Android, iOS, Raspberry Pi, Arduino, NXP, Electron, Xcode, QT, CATIA, NI Multisim, NX, LabView
Language:	C, C++, Bash, Python [Tensorflow, OpenCV, Numpy, Scikitlearn], Obj C, Java, avaScript [Node, React, D3], HTML5/CSS3, Matlab
Protocols:	CAN, SPI, UART, GPS, Bluetooth, TCP, UDP, HTTP, FTP, SSH, Bitcoin, Bittorrent